Machine Learning Based Analog Circuit Fault Diagnosis

PDF 2 — Chapters 5-6: Electronics Fundamentals & LTspice

RC/RL/RLC Circuit Theory | LTspice Simulation Software

CHAPTER 5 — Analog Electronics Fundamentals

Chapter 5 — Analog Electronics Fundamentals (The Electronics Behind Our Project) Before Starting Till now we have talked about Machine Learning. Now let's forget ML for a while. Because before understanding how the computer finds faults, you should first understand how the circuit behaves. Remember

Machine Learning

doesn't create the waveform. The circuit does.

What you will know after finishing this chapter

After finishing this chapter you should be able to answer [OK] What is an analog circuit? [OK] What are RC, RL and RLC circuits? [OK] Why do these circuits produce different waveforms? [OK] Why do faults change waveforms? [OK] Why do resistors, capacitors and inductors affect transient response? [OK] Explain the electronics behind the project.

Difficulty

Basics 5/5

Theory 5/5

Coding 1/5

Viva Importance 5/5

Let's Begin

What exactly is an analog circuit? Simple definition. An analog circuit is an electronic circuit whose voltage and current can take any value within a range. Example Voltage 0V 1.25V 2.84V 4.62V 3.19V. Notice Values are continuous. Not just 0 or

PROFESSOR ASKS

What is an analog circuit?

EXCELLENT ANSWER

An analog circuit processes continuously varying electrical signals, unlike digital circuits which operate using discrete logic levels. Digital vs Analog Let's compare. Digital **OFF** — 0. **ON** — 1. Only two values. Analog 0.15 0.83 2.47 3.16 4.82. Infinite possibilities. Question. Which one is harder to diagnose? Obviously Analog. Why Did We Choose Analog Circuits? Because fault diagnosis is much harder. Suppose a resistor changes from

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100 Ω
```

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110 Ω
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Question. Will the circuit stop working? Maybe not. Instead the waveform changes slightly. Detecting that small change is difficult. That's why

Machine Learning

becomes useful.

Components Used

Our circuits mainly use three passive components.

1. Resistor

Symbol /\/\/\ Purpose Controls current. Think of water flowing through a pipe. Question. What happens if the pipe becomes narrower? Water flow decreases. Same idea. Resistor limits current.

PROFESSOR ASKS

What does a resistor do?

EXCELLENT ANSWER

A resistor limits current flow and creates voltage drops according to Ohm's Law. Ohm's Law One of the most important equations in electronics.

V = I × R

where **V** — Voltage. **I** — Current. **R** — Resistance. Remember this forever. Question. Suppose Resistance increases. Current? Using

I = V / R

Current decreases. Simple.

2. Capacitor

Symbol

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Question. What does a capacitor do? Think of a water tank. Water doesn't fill instantly. It takes time. Similarly a capacitor stores electrical charge.

PROFESSOR ASKS

What is a capacitor?

EXCELLENT ANSWER

A capacitor stores electrical energy in the form of an electric field and opposes sudden changes in voltage. Remember the last line. Very important. Question. Can capacitor voltage change instantly? No. Capacitors hate sudden voltage changes. Remember this sentence.

3. Inductor

Symbol (((((Question. What does an inductor do? Think of a heavy flywheel. It doesn't like sudden speed changes. Similarly an inductor doesn't like sudden current changes.

PROFESSOR ASKS

What is an inductor?

EXCELLENT ANSWER

An inductor stores energy in a magnetic field and opposes sudden changes in current. Again remember the last line.

MEMORY TRICK

__Capacitor__ — Opposes. Voltage Change __Inductor__ — Opposes. Current Change Very useful. Why These Components Matter Question. Why are we discussing resistors, capacitors and inductors? Because our waveforms depend on them. If one component changes, the waveform changes. Machine Learning learns those waveform changes. RC Circuit Now our first circuit. Question. What does RC mean? Simple. __R__ — Resistor. __C__ — Capacitor. __Together__ — RC Circuit. Suppose you apply 5V. Question. Will capacitor voltage immediately become 5V? No. It slowly rises. Example Voltage 5V | ______ | / | / | / | / 0V |_____/________________ Time Notice Slow rise. This is called Charging. Question. Why slow? Because capacitor needs time to store charge. Professor Why doesn't the capacitor charge instantly? Excellent Answer Because a capacitor stores charge gradually. The charging rate is controlled by the resistance and capacitance values of the circuit. Excellent. Time Constant Very important. One of the most common viva questions. Formula τ = RC Read it as Tau. Question. What is Time Constant? Simple. It tells us how fast the capacitor charges or discharges. Question. Suppose Resistance increases. Time Constant? τ = RC Resistance ↑ — Time Constant. ↑ Charging becomes slower. Question. Suppose Capacitance increases. Again Time Constant increases. Again charging becomes slower. Professor What happens if resistance doubles? Excellent Answer The RC time constant doubles, causing the capacitor to charge and discharge more slowly. RL Circuit Now replace capacitor with inductor. Question. What happens? Current doesn't increase instantly. Instead it rises gradually. Reason? Inductor opposes current changes. Formula τ = L / R Notice different formula. __RC__ — RC. __RL__ — L/R. Remember. Professor What is the RL time constant? Excellent Answer The RL time constant is the ratio of inductance to resistance and determines how quickly current changes in the circuit. RLC Circuit Now combine all three. R + L + C Question. What happens? The waveform becomes more interesting. Sometimes it oscillates. Example Voltage | | /\ | / \ | / \ |__/___________ Time Sometimes the oscillation gradually disappears. This is called Damping. Professor Why does an RLC circuit oscillate? Excellent Answer Because energy continuously exchanges between the capacitor's electric field and the inductor's magnetic field while resistance gradually dissipates energy. Excellent. Why Faults Change Waveforms This is the most important part of this chapter. Question. Suppose Resistor increases. Will waveform remain the same? No. Rise Time changes. Peak changes. Settling Time changes. Energy changes. Everything changes. Question. Suppose Capacitor becomes open. Again waveform changes. Question. Suppose Inductor becomes short. Again waveform changes. Now Question. Can Machine Learning see the resistor? No. Can it see the capacitor? No. It only sees the waveform. That is why waveform contains all the information. One Beautiful Analogy Think of a human. Question. Can doctor see virus directly? Usually No. Doctor looks at symptoms. Example Fever. Cough. Weakness. Similarly Machine Learning cannot see the resistor. It sees the symptoms. __Symptoms__ — Waveform Features. Beautiful analogy. Use this in viva. Why Do We Extract Features? Now this question makes sense. Waveform changes because components change. Features measure those changes. Example __Rise Time__ — Capacitor effect. __Peak Voltage__ — Circuit response. __DC Level__ — Steady State. __Energy__ — Overall signal strength. Now Machine Learning can distinguish faults. Common Professor Question How does Machine Learning know that a resistor is faulty? Excellent Answer Machine learning does not directly identify the resistor. It learns patterns in the waveform features that are associated with different fault conditions during training. Excellent. Another Professor Question Can two different faults produce similar waveforms? Yes. Very important. That is exactly why classification is difficult. Sometimes RC Open and another fault produce very similar waveforms. That is why we needed better features like DC Level. Notice everything is now connecting.

COMMON MISTAKES

MISTAKE: [X] Capacitor opposes current. Wrong. Capacitor opposes voltage change.

MISTAKE: [X] Inductor opposes voltage. Wrong. Inductor opposes current change.

MISTAKE: [X] Machine Learning detects components. Wrong. Machine Learning detects patterns in waveforms.

REVISION NOTES

Remember
these five things.
Resistor — Limits Current.
Capacitor — Stores Charge.
Opposes Voltage Change.
Inductor — Stores Magnetic Energy.
Opposes Current Change.
Fault — Changes Waveform.
Machine Learning — Learns Waveform Changes.

MEMORY TRICK

Remember this: R → Controls Current C → Controls Voltage Change L → Controls Current Change Or even shorter: RCL R = Resistance limits current. C = Capacitor resists voltage changes. L = Inductor resists current changes. Knowledge Check Level 1 What is an analog circuit? What does a resistor do? What does a capacitor do? What does an inductor do? Why are analog circuits difficult? Level 2 Why does a capacitor charge slowly? What is the RC time constant? What is the RL time constant? Why does an RLC circuit oscillate? Why do faults change waveforms? Professor Level Why is analog fault diagnosis more challenging than digital fault diagnosis? Explain how a change in resistance affects the transient response of an RC circuit. Why can two different faults produce similar waveforms? How does your machine learning model infer the presence of a fault without directly measuring component values? Why is understanding circuit behavior important before applying machine learning? End of Chapter 5

Congratulations. This is the first chapter where the electronics and machine learning truly connect. From now on, whenever we discuss a feature like Rise Time, Peak Voltage, or DC Level, you'll understand that these aren't just numbers—they are direct consequences of how resistors, capacitors, and inductors influence the circuit's transient response.

One Suggestion for Improving the Bible While writing this chapter, I noticed something. Starting from the next chapter, I'll add a small section called: Connection to Our Project At the end of each chapter, we'll explicitly answer: "How is everything we learned in this chapter actually used in our project?" That way, every concept immediately ties back to your implementation instead of feeling like separate theory. I think this will make revision much easier before your viva.

CHAPTER 6 — LTspice (The Source of Our Entire Dataset)

Chapter 6 — LTspice (The Source of Our Entire Dataset) Before Starting Till now we have learned [OK] What is the project? [OK] Why it was built?

[OK] Objectives

[OK] Methodology

[OK] Analog Electronics

Now we reach the first software used in our project. Without this software our project would not exist. Because without LTspice there is no dataset. Without dataset there is no Machine Learning. So, LTspice is the foundation of our entire project.

What you will know after finishing this chapter

After finishing this chapter you should be able to answer [OK] What is LTspice? [OK] Why did we use LTspice? [OK] Why not hardware? [OK] What is simulation? [OK] What is transient analysis? [OK] How did we generate the dataset? [OK] Why is LTspice important in our project?

Difficulty

Basics 5/5

Theory 5/5

Coding 1/5

Viva Importance 5/5

Let's Begin

What is LTspice? Simple answer. LTspice is an electronic circuit simulator. Question. What does simulator mean? It means instead of building the circuit physically, we build it inside a computer. Think like this. Instead of buying Resistor Capacitor Inductor

Power Supply

we simply drag and connect them on the screen. Then LTspice behaves exactly like the real circuit.

PROFESSOR ASKS

What is LTspice?

EXCELLENT ANSWER

LTspice is a SPICE-based electronic circuit simulation software used to analyze the behavior of analog electronic circuits without physically building them. Remember the keyword

Simulation Software

What is Simulation? Question. Suppose I ask you to study an RC circuit.

Method 1

Buy components. Build circuit. Connect oscilloscope. Measure voltage.

Method 2

Open LTspice. Draw circuit. Click Run. Observe waveform. Which is easier? Obviously Method 2. That is Simulation.

Real Life Analogy

Suppose you want to become a pilot. Question. Do airlines directly give you a real airplane? No. They first train you using a flight simulator. Why? Because simulation is safer cheaper faster LTspice is exactly that simulator for electronics. Why Did We Use LTspice? Question. Why not build real circuits?

Good question. Let's compare. Hardware Need Components Breadboard Oscilloscope Power Supply Multimeter Expensive. Time consuming. Component failures are difficult to reproduce. LTspice Need only a computer. Can simulate thousands of circuits. Very fast. Repeatable. Easy.

PROFESSOR ASKS

Why did you use LTspice instead of hardware?

EXCELLENT ANSWER

LTspice allows accurate analog circuit simulation under controlled conditions. It enables rapid dataset generation, systematic fault injection, and reproducible experiments without requiring physical hardware. Excellent answer.

One Important Thing

Never say LTspice is better than hardware. Wrong. Instead say LTspice is appropriate for this stage of the project. Very professional. Can LTspice Replace Hardware? Question. Does simulation mean hardware is useless? No. Simulation helps us develop the methodology. Hardware is required for real-world validation. Remember this.

PROFESSOR ASKS

Are simulation results identical to hardware?

EXCELLENT ANSWER

No. Real hardware introduces additional factors such as measurement noise, component tolerances, temperature variations, and parasitic effects. Therefore, hardware validation is an important future extension. Very important. What Did We Simulate? Question. What exactly did we simulate? Not random circuits. We simulated our analog circuits. Example RC Circuit RL Circuit RLC Circuit Each circuit had Healthy and Faulty versions.

Healthy Circuit

Example

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Resistor = 100 Ω
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Capacitor = 1 μF Everything works properly.

Faulty Circuit

Example

Resistor Open

or

Capacitor Short

or Resistance changed Again simulate. Again collect waveform.

PROFESSOR ASKS

How did you generate different fault classes?

EXCELLENT ANSWER

Different fault classes were generated by intentionally modifying circuit components to represent healthy and faulty operating conditions before running the simulations. Notice the words Intentionally modifying This is called Fault Injection. We'll study it later. What Happens After Clicking Run? Question. Suppose we click Run. Question. What does LTspice calculate? It solves the electrical equations of the circuit. Then it generates Voltage Current Power and many other signals. In our project we mainly use Voltage Waveforms. What is a Waveform? Suppose the capacitor starts charging. Voltage doesn't jump like this **0** — 5V. Instead it changes gradually. Example Voltage

```
5V | _______
```

| / | / | /

```
0V |_____/____________ Time
```

This graph is called a Waveform.

PROFESSOR ASKS

What is a waveform?

EXCELLENT ANSWER

A waveform is the graphical representation of how an electrical quantity such as voltage changes with time. Simple. Why Voltage? Question. Why did we use Voltage? Why not Current? Good question. Voltage waveforms contain enough information to distinguish the fault conditions used in our study. Current could also be analyzed in future work.

PROFESSOR ASKS

Why voltage instead of current?

EXCELLENT ANSWER

Voltage was selected because it captures the transient behavior required for fault classification while remaining straightforward to obtain consistently across all simulated circuits.

Transient Analysis

This is one of the most important terms in the project. Question. What is Transient? Imagine switching ON a circuit. Immediately after switching, everything starts changing. Voltage rises. Current changes. After some time everything settles. That changing period is called the Transient Response.

PROFESSOR ASKS

What is transient analysis?

EXCELLENT ANSWER

Transient analysis studies how circuit voltages and currents change with time immediately after an input or switching event before reaching steady state. Excellent. Why Did We Use Transient Analysis? Question. Why not DC Analysis? Because faults change the transient response. Features like Rise Time Settling Time Peak Voltage all come from Transient Analysis. Without transient analysis our project would not exist. Complete LTspice Workflow Everything inside LTspice can be summarized using one diagram.

Design Circuit

Assign Component Values

Introduce Fault

Run Transient Analysis

Observe Waveform

Export Data

Congratulations. You now understand everything LTspice does in our project. Where Does the Dataset Come From? Question. Suppose LTspice shows the waveform. How does Python receive it? Simple. We export the waveform as CSV. Example Time,Voltage 0,0 0.001,0.52 0.002,1.08. Now Python can read it.

PROFESSOR ASKS

Why CSV?

EXCELLENT ANSWER

CSV is lightweight, easy to export, portable across software, and can be directly processed using Python libraries such as Pandas.

Connection to Our Project

Everything we have learned now connects to our project.

Analog Circuit

LTspice

Transient Analysis

Voltage Waveform

CSV Export — Python.

Machine Learning

Without LTspice there would be no waveform. Without waveform there would be no features. Without features there would be no SVM. Everything starts here.

COMMON MISTAKES

MISTAKE: [X] LTspice generates Machine Learning data. Wrong. LTspice generates electrical waveforms. Python converts them into ML data.

MISTAKE: [X] Simulation is always identical to hardware. Wrong. Simulation is an approximation. Hardware validation is still important.

MISTAKE: [X] Transient Analysis measures only voltage. Wrong. It can calculate many electrical quantities. We selected Voltage for our project.

REVISION NOTES

Remember
these six points.
LTspice — Circuit Simulator.
Simulation — Virtual Experiment.
Transient Analysis — Voltage vs Time.
Waveform — Project Input.
CSV — Bridge between LTspice and Python.
Everything
starts
with LTspice.

MEMORY TRICK

Remember this sequence: D → F → R → W → E Expand it as: D = Design Circuit F = Fault Injection R = Run Simulation W = Waveform E = Export CSV Whenever someone asks "How did you generate the dataset?" Just remember D → F → R → W → E

KNOWLEDGE CHECK

Level 1

- What is LTspice?
- What is simulation?
- What is a waveform?
- What is transient analysis?
- Why did we use LTspice?

Level 2

- Why not use real hardware?
- Why transient analysis instead of DC analysis?
- Why export data as CSV?
- How does Python receive the waveform?
- Why is LTspice the foundation of the project?

Professor Level

- What assumptions does LTspice make during simulation?
- What limitations of simulation could affect your dataset?
- Why is transient analysis particularly suitable for fault diagnosis?
- Could your methodology be extended to hardware measurements? What changes would be required?
- If current waveforms were used instead of voltage waveforms, how might the project change?
- End of Chapter 6

Congratulations. You now understand the software foundation of the project. More importantly, you understand that LTspice is not just a simulator—it is the data source for the entire machine learning pipeline.

Author's Note One thing I want to improve starting from the next chapter is to include actual snippets from your project wherever possible. For example, when we discuss dataset generation, we'll connect the theory directly to your implementation, your folder structure, and the code you wrote. That will make the Bible even more personalized and much closer to your actual project rather than a generic guide.